Optimization

Francesca Maggioni

Lesson 1

Department of Management, Information and Production Engineering
University of Bergamo

The Instructor

Francesca Maggioni

- Professor of Operations Research (ssd MATH-06/A)

  Department of Management, Information & Production Engineering - University of Bergamo
- E-mail: `francesca.maggioni@unibg.it`
- Home page: `http://www.francescamaggioni.it`
- Office hours: **Monday 2:30-3:30 pm**

Giovanni Micheli (Tutor)

- E-mail: giovanni.micheli@unibg.it

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UNIVERSITÀ DEGLI STUDI DI BERGAMO

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Course Organization

Lessons:

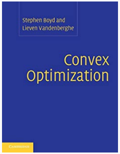
- Tuesday: 3:30 PM – 6:30 PM

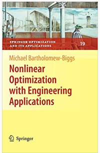
Materials and Announcements:

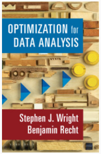
- https://elearning15.unibg.it/

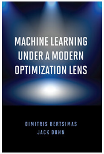
References

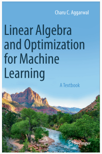
- Stephen Boyd & Lieven Vandenberghe (2004) *Convex Optimization*, Cambridge University Press.
- Michael Bartholomew-Biggs (2008) *Nonlinear Optimization with Engineering Applications*, Springer.
- Stephen J. Wright & Benjamin Recht (2022) *Optimization for Data Analysis*, Cambridge University Press.
- Dimitris Bertsimas and Jack Dunn (2019) *Machine Learning Under a Modern Optimization Lens*, Dynamic Ideas LLC.
- Charu C. Aggarwal Linear Algebra and Optimization for Machine Learning.











- Lecture notes and further reading will be posted to the **e-learning platform**.

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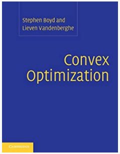

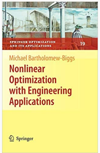

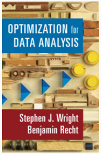

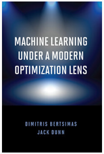

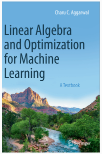

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Assessment Method and Criteria

The **final exam** consists in two parts:

- Oral discussion about applied assignments and case studies (50% of the final grade). Material for the exam project will be available in the e-learning platform. Students may work in small groups or individually.
- Oral exam on the theory (50% of the final grade).

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- Oral exam on the theory (50% of the final grade).

Outline

- **Introduction to non linear optimization**
  - Constrained (and unconstrained) optimization
  - Convex optimization
- **Algorithms for non linear optimization**
  - Gradient Methods
  - Subgradient Methods
  - Stochastic Gradient Descent
  - Coordinate Descent
  - Second-Order methods including Newton and Quasi-Newton
  - Derivative-free optimization
- **Optimization problems for machine learning**
  - Regression
  - Clustering
  - Density estimation
  - Classification: Support vector machine models
  - Classification: Decision tree via mixed integer linear programming
  - Neural network training.

Motivation of the course

Optimization is everywhere:

- Machine learning, big data, statistics, data analysis of all kinds, finance, logistics, planning, control theory, mathematics, search engines, simulations, and many other applications.
- **Mathematical Modeling**: defining & modeling the optimization problem.
- **Computational Optimization**: running an (appropriate) optimization algorithm.

Optimization for Machine Learning:

- **Mathematical Modeling**: defining & and measuring the machine learning model.
- **Computational Optimization**: learning the model parameters.
- Theory *vs.* practice:
  - libraries are available, algorithms treated as black box by most practitioners;
  - Not here: we look inside the algorithms and try to understand why and how fast they work!

- Main approaches:
  - Gradient Descent.
  - Stochastic Gradient Descent (SGD).
  - Coordinate Descent.
- History:
  - 1847: Cauchy proposes gradient descent.
  - 1950s: Linear Programs, soon followed by non-linear, SGD.
  - 1980s: General optimization, convergence theory.
  - 2005-today: Large scale optimization, convergence of SGD.

Non Linear Optimization

- A **non linear programming problem (NLP)**, is a mathematical programming problem of the kind:

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ s.t. \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \ldots, m \\ & \mathbf{x} \geq \mathbf{0} \end{aligned} \tag{NLP}$$

  where $f : dom(f) \subseteq \mathbb{R}^n \to \mathbb{R}$ and $g_i : dom(g_i) \subseteq \mathbb{R}^n \to \mathbb{R}$ are given functions and at least one of these functions (objective function or constraints) is non linear.

Example

$$\begin{aligned} \max \quad & \left(x_1 - \tfrac{1}{2}\right)^2 + \left(x_2 - \tfrac{1}{2}\right)^2 \\ s.t. \quad & x_1 + x_2 \geq 1, \quad x_1 \leq 1 \end{aligned}$$

No algorithm that will solve *every* specific problem fitting this format is available. However, substantial progress has been made for some important special cases of this problem by making various assumptions about these functions.

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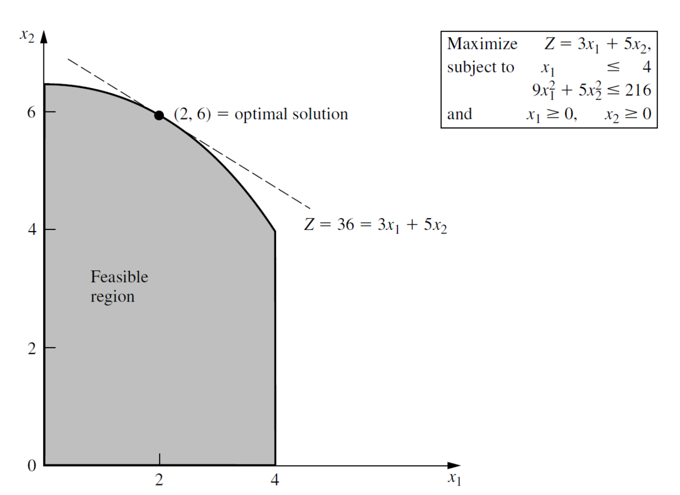
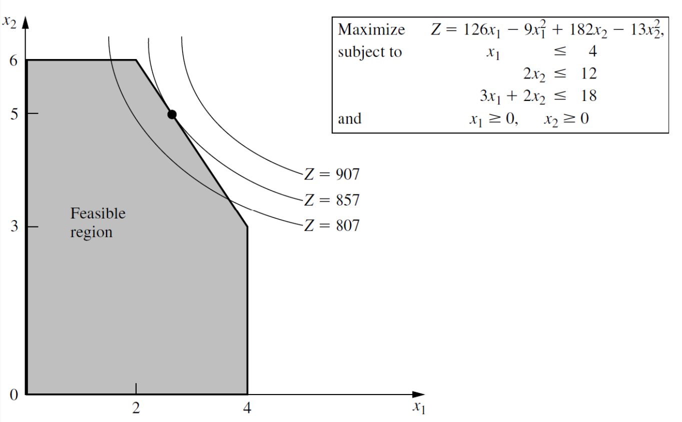
Graphical illustration of non linear programming problems

- The optimal solution still lies on the boundary of the feasible region. However, it **is not a vertex** as in LP!

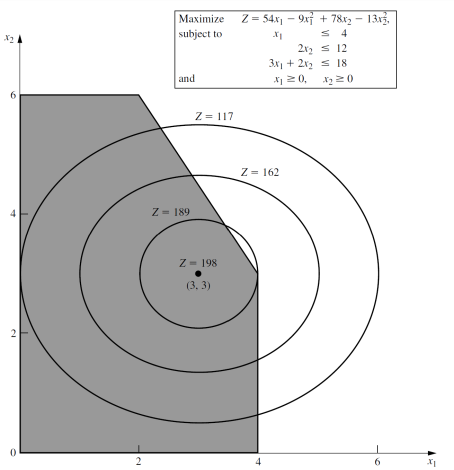


- Non linear objective function subject to linear constraints.
- Optimal solution on the boundary: $(\frac{8}{3}, 5)$



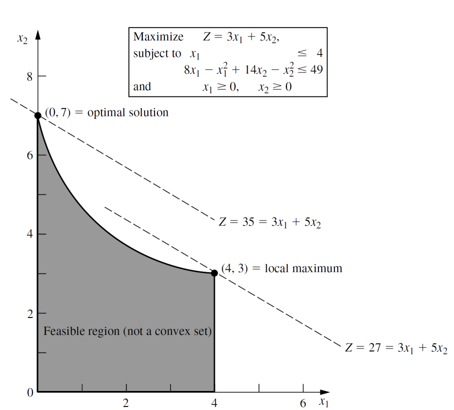
Graphical illustration of non linear programming problems

- Optimal solution inside the feasible region: $(3, 3)$.



A general algorithm for solving similar problems needs to consider all solutions in the feasible region, not just those on the boundary.

Graphical illustration of non linear programming problems



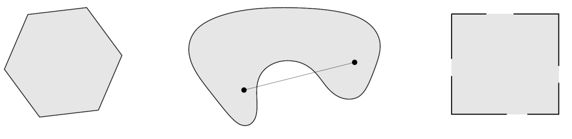
- The feasible region is not a convex set. Consequently, we cannot guarantee that a local maximum is a global maximum. In fact, this example has two local maxima, (0, 7) and (4, 3), but only (0, 7) is a global maximum. To guarantee that a local maximum is a global maximum for a nonlinear programming problem we need to consider a **convex programming problem**.

Convex Optimization

Convex Sets

A set C is **convex** if the line segment between any two points of C lies in C, *i.e.*, if for any $\mathbf{x}, \mathbf{y} \in C$ and any λ with $0 \leq \lambda \leq 1$, we have:

$$\lambda \mathbf{x} + (1 - \lambda)\mathbf{y} \in C.$$



- *Left*: Convex.
- *Middle*: Not convex, since line segment not in set.
- *Right*: Not convex, not all boundary points are contained in the set.

Properties of Convex Sets

- Intersections of convex sets are convex.
  - Let $C_i, i \in I$ be convex sets, where I is a (possibly infinite) index set. Then $C = \cap_{i \in I} C_i$ is a convex set.
- Projections onto convex sets are unique, and often efficient to compute.

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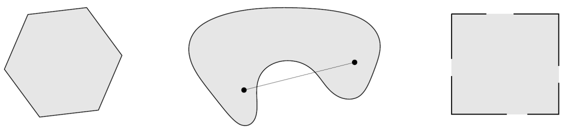

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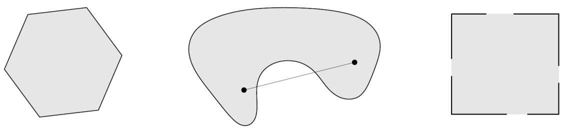

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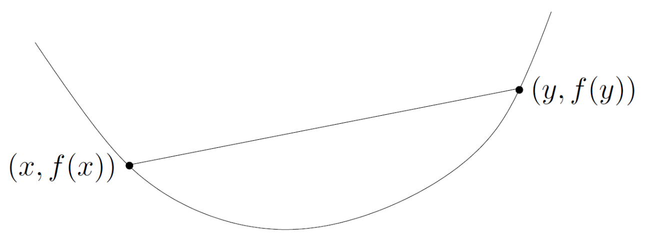
Convex Functions

A function $f : \mathbb{R}^d \to \mathbb{R}$ is **convex** if

1. $\text{dom}(f)$ is a convex set;
2. for all $\mathbf{x}, \mathbf{y} \in \text{dom}(f)$ and any λ with $0 \leq \lambda \leq 1$, we have:

$$f(\lambda\mathbf{x} + (1-\lambda)\mathbf{y}) \leq \lambda f(\mathbf{x}) + (1-\lambda)f(\mathbf{y}).$$



Geometrically: The line segment between $(\mathbf{x}; f(\mathbf{x}))$ and $(\mathbf{y}; f(\mathbf{y}))$ lies above the graph of f.

Convex Optimization Problems are of the form:

$$\begin{aligned} \min \quad & f(\mathbf{x}) \\ s.t. \quad & \mathbf{x} \in X \end{aligned}$$

where both:

- f is a convex function;
- $X \subset \mathrm{dom}(f)$ is a convex set (note: $\mathbb{R}^d$ is convex).

Crucial Property of Convex Optimization Problems

Every **local minimum** is a **global minimum**.

Examples of convex functions:

- Linear functions: $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$.
- Affine functions: $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$.
- Exponential: $f(\mathbf{x}) = e^{\alpha \mathbf{x}}$.
- Norms. Every norm on $\mathbb{R}^d$ is convex.

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Convexity of a norm $||\mathbf{x}||$

By the triangle inequality:

$$||\mathbf{x}+\mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}||$$

and homogeneity of a norm:

$$||a\mathbf{x}|| = |a| \cdot ||\mathbf{x}||$$

with a scalar, we have:

$$||\lambda\mathbf{x} + (1-\lambda)\mathbf{y}|| \leq ||\lambda\mathbf{x}|| + ||(1-\lambda)\mathbf{y}|| = \lambda||\mathbf{x}|| + (1-\lambda)||\mathbf{y}||.$$

We used the triangle inequality for the inequality and homogeneity for the equality.

Remark

For convex optimization problems, all algorithms (Coordinate Descent, Gradient Descent, Stochastic Gradient Descent, Projected and Proximal Gradient Descent) do converge to the global optimum!

Example Theorem

The **convergence rate** is proportional to $\frac{1}{t}$, *i.e.*:

$$f(\mathbf{x}_t) - f(\mathbf{x}^*) \leq \frac{c}{t}$$

where $\mathbf{x}^*$ is some optimal solution to the problem.

Meaning: Approximation error converges to 0 over time.

Example of results obtained in the course

f	Algorithm	Rate	# Iter	Cost/iter
non-smooth	center of gravity	$\exp\left(-\frac{t}{n}\right)$	$n \log\left(\frac{1}{\varepsilon}\right)$	1 ∇, 1 n-dim $\int$
non-smooth	ellipsoid method	$\frac{R}{r}\exp\left(-\frac{t}{n^2}\right)$	$n^2 \log\left(\frac{R}{r\varepsilon}\right)$	1 ∇, mat-vec $\times$
non-smooth	Vaidya	$\frac{Rn}{r}\exp\left(-\frac{t}{n}\right)$	$n \log\left(\frac{Rn}{r\varepsilon}\right)$	1 ∇, mat-mat $\times$
quadratic	CG	exact $\exp\left(-\frac{t}{\kappa}\right)$	n $\kappa \log\left(\frac{1}{\varepsilon}\right)$	1 ∇
non-smooth, Lipschitz	PGD	$RL/\sqrt{t}$	R^2L^2/ε^2	1 ∇, 1 proj.
smooth	PGD	$\beta R^2/t$	$\beta R^2/\varepsilon$	1 ∇, 1 proj.
smooth	AGD	$\beta R^2/t^2$	$R\sqrt{\beta/\varepsilon}$	1 ∇
smooth (any norm)	FW	$\beta R^2/t$	$\beta R^2/\varepsilon$	1 ∇, 1 LP
strong. conv., Lipschitz	PGD	$L^2/(\alpha t)$	$L^2/(\alpha\varepsilon)$	1 ∇ , 1 proj.
strong. conv., smooth	PGD	$R^2\exp\left(-\frac{t}{\kappa}\right)$	$\kappa \log\left(\frac{R^2}{\varepsilon}\right)$	1 ∇ , 1 proj.
strong. conv., smooth	AGD	$R^2\exp\left(-\frac{t}{\sqrt{\kappa}}\right)$	$\sqrt{\kappa}\log\left(\frac{R^2}{\varepsilon}\right)$	1 ∇
$f+g$, f smooth, g simple	FISTA	$\beta R^2/t^2$	$R\sqrt{\beta/\varepsilon}$	1 ∇ of f Prox of g
$\max_{y\in\mathcal{Y}} \varphi(x,y)$, φ smooth	SP-MP	$\beta R^2/t$	$\beta R^2/\varepsilon$	MD on $\mathcal{X}$ MD on $\mathcal{Y}$
linear, $\mathcal{X}$ with F ν-self-conc.	IPM	$\nu\exp\left(-\frac{t}{\sqrt{\nu}}\right)$	$\sqrt{\nu}\log\left(\frac{\nu}{\varepsilon}\right)$	Newton step on F
non-smooth	SGD	$BL/\sqrt{t}$	B^2L^2/ε^2	1 stoch. ∇, 1 proj.
non-smooth, strong. conv.	SGD	$B^2/(\alpha t)$	$B^2/(\alpha\varepsilon)$	1 stoch. ∇, 1 proj.
$f=\frac{1}{m}\sum f_i$ f_i smooth strong. conv.	SVRG	–	$(m+\kappa)\log\left(\frac{1}{\varepsilon}\right)$	1 stoch. ∇

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Lemma

Suppose that dom(f) is open and that f is differentiable; in particular, the **gradient** (vector of partial derivatives)

$$\nabla f(\mathbf{x}) := \left(\frac{\partial f}{\partial x_1}(\mathbf{x}), \ldots, \frac{\partial f}{\partial x_d}(\mathbf{x})\right)$$

exists at every point $\mathbf{x} \in$ dom(f). Then f is convex if and only if dom(f) is convex and

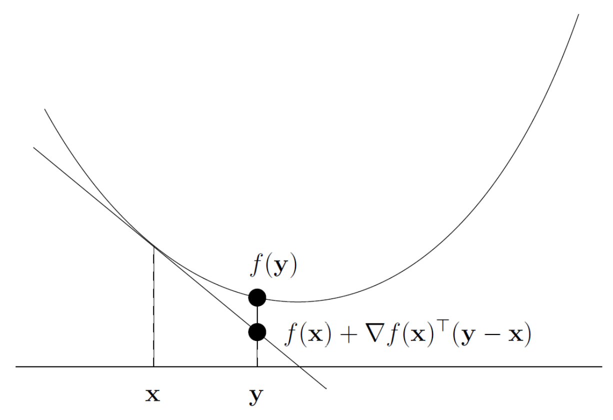
$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x})$$

holds for all $\mathbf{x}, \mathbf{y} \in$ dom(f).

First-order Characterization of Convexity

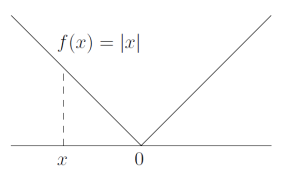
$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}), \quad \mathbf{x}, \mathbf{y} \in \mathrm{dom}(f).$$

Graph of f is above all its tangent hyperplanes.

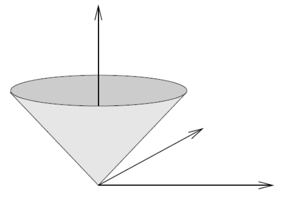


Non-differentiable Functions

Non-differentiable functions are also relevant in practice:



More generally, $f(\mathbf{x}) = ||\mathbf{x}||$ (Euclidean norm). For $d = 2$, graph is the ice cream cone:



Lemma

Suppose that $\mathrm{dom}(f)$ is open and that f is twice differentiable; in particular, the **Hessian** (matrix of second partial derivatives)

$$\nabla^2 f(\mathbf{x}) := \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{x}) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_d}(\mathbf{x}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_2 \partial x_2}(\mathbf{x}) & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_d}(\mathbf{x}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1}(\mathbf{x}) & \frac{\partial^2 f}{\partial x_d \partial x_2}(\mathbf{x}) & \dots & \frac{\partial^2 f}{\partial x_d \partial x_d}(\mathbf{x}) \end{bmatrix}$$

exists at every point $\mathbf{x} \in \mathrm{dom}(f)$ and is symmetric. Then f is convex if and only if $\mathrm{dom}(f)$ is convex, and for all $\mathbf{x} \in \mathrm{dom}(f)$, we have

$$\nabla^2 f(\mathbf{x}) \succeq 0$$

i.e., $\nabla^2 f(\mathbf{x})$ is positive semidefinite.

Definition

Let M be a symmetric matrix. M is definite if the sign of $\mathbf{x}^\top M\mathbf{x}$ does not change, for all $\mathbf{x}$. If not, then M is indefinite. Specifically

- *M is positive semidefinite if $\mathbf{x}^\top M\mathbf{x} \geq 0$ for all $\mathbf{x}$;*
- *M is positive definite if $\mathbf{x}^\top M\mathbf{x} \geq 0$ for all $\mathbf{x} \neq 0$;*
- *M is negative semidefinite if $\mathbf{x}^\top M\mathbf{x} \leq 0$ for all $\mathbf{x}$;*
- *M is negative definite if $\mathbf{x}^\top M\mathbf{x} \leq 0$ for all $\mathbf{x} \neq 0$.*

Theorem

- *A symmetric matrix* M *is* **positive semidefinite** *if and only if the determinant of each principal submatrix (not only North-West) is non negative and the determinant of M is null.*
- *A symmetric matrix* M *is* **positive definite** *if and only if the determinant of all the North-West submatrices are* > 0*.*
- *A symmetric matrix* M *is* **negative semidefinite** *if and only if the determinant of each principal submatrix (not only North-West) of odd order is non positive, the determinant of each principal submatrix (not only North-West) of even order is non negative and the determinant of M is null.*
- *A symmetric matrix* M *is* **negative definite** *if and only if the determinant of all the North-West submatrices of odd order are* < 0 *and the determinant of all the North-West submatrices of even order are* > 0*.*

Second-order Characterization of Convexity

Example 1

Let A be the following matrix

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

Its **North-West submatrices** of order 1, 2 and 3 are, respectively

$$\begin{bmatrix} [1] & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 4 \end{bmatrix} \quad \begin{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} & \begin{matrix} 1 \\ 1 \end{matrix} \\ \begin{matrix} 1 & 1 \end{matrix} & 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

Given that $\det[1] = 1 > 0$, $\det \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = 1 > 0$ and $\det A = 3 > 0$, A is **positive definite** because the determinants of all its North-West submatrices are > 0.

Exercises

Determine if the following hessians are positive/negative semidefinite/definite:

$$B = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & -1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} -1 & -1 & -1 \\ -1 & -2 & -1 \\ -1 & -1 & -3 \end{bmatrix}$$